

B6 Polynomials

Name: _____

Class: _____

Date: _____

Time: **107 minutes**

Marks: **85 marks**

Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

Simplify the algebraic fraction $\frac{2x^2 + x - 3}{x^2 - 1}$

(Total 3 marks)

Q2.

The polynomial $p(x)$ is given by

$$p(x) = x^3 + x^2 - 10x + 8$$

- (a) (i) Using the factor theorem, show that $x - 2$ is a factor of $p(x)$.

(2)

- (ii) Hence express $p(x)$ as the product of three linear factors.

(3)

- (b) Sketch the curve with equation $y = x^3 + x^2 - 10x + 8$, showing the coordinates of the points where the curve cuts the axes.

(You are not required to calculate the coordinates of the stationary points.)

(4)

(Total 9 marks)

Q3.

The polynomial $f(x)$ is defined by $f(x) = 3x^3 + 2x^2 - 7x + 2$.

- (a) Find $f(1)$.

(1)

- (b) Show that $f(-2) = 0$.

(1)

- (c) Hence, or otherwise, show that

$$\frac{(x-1)(x+2)}{3x^3 + 2x^2 - 7x + 2} = \frac{1}{ax+b}$$

where a and b are integers.

(3)

(Total 5 marks)

Q4.

The polynomial $p(x)$ is given by $p(x) = x^3 - 4x^2 + 3x$.

- (a) Use the Factor Theorem to show that $x - 3$ is a factor of $p(x)$.

(2)

- (b) Express $p(x)$ as the product of three linear factors.

(2)

(Total 4 marks)

Q5.

- (a) The polynomial $p(x)$ is defined by $p(x) = 6x^3 - 19x^2 + 9x + 10$.

- (i) Find $p(2)$.

(1)

- (ii) Use the Factor Theorem to show that $(2x + 1)$ is a factor of $p(x)$.

(3)

- (iii) Write $p(x)$ as the product of three linear factors.

(2)

- (b) Hence simplify

$$\frac{3x^2 - 6x}{6x^3 - 19x^2 + 9x + 10}$$

(2)

(Total 8 marks)

Q6.

- (a) The function f is defined for all values of x by $f(x) = x^3 - 3x^2 - 6x + 8$.

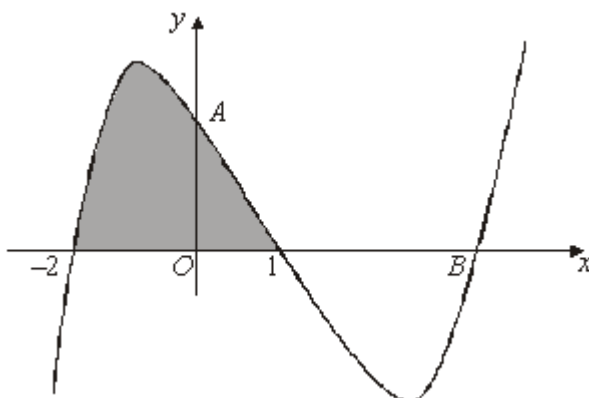
- (i) Given that $f(1) = 0$ and $f(-2) = 0$, write down two linear factors of $f(x)$.

(2)

- (ii) Hence express $x^3 - 3x^2 - 6x + 8$ as the product of three linear factors.

(2)

- (b) The curve with equation $y = x^3 - 3x^2 - 6x + 8$ is sketched below.



- (i) The curve intersects the y -axis at the point A . Find the y -coordinate of A . (1)
- (ii) The curve crosses the x -axis when $x = -2$, when $x = 1$ and also at the point B . Use the results from part (a) to find the x -coordinate of B . (1)
- (c) (i) Find $\int (x^3 - 3x^2 - 6x + 8) dx$ (4)
- (ii) Hence find the area of the shaded region bounded by the curve and the x -axis. (3)
- (Total 13 marks)**

Q7.

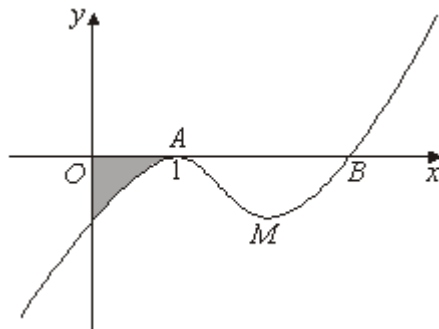
- (a) Simplify $(k + 5)^2 - 12k(k + 2)$. (2)
- (b) The quadratic equation $3(k + 2)x^2 + (k + 5)x + k = 0$ has real roots.
- (i) Show that $(k - 1)(11k + 25) \leq 0$. (5)
- (ii) Hence find the possible values of k . (3)
- (Total 10 marks)**

Q8.

- (a) (i) Using the binomial expansion, or otherwise, express $(2 + x)^3$ in the form $8 + ax + bx^2 + x^3$, where a and b are integers. (3)
- (ii) Write down the expansion of $(2 - x)^3$. (2)
- (b) Hence show that $(2 + x)^3 - (2 - x)^3 = 24x + 2x^3$. (2)
- (c) Hence show that the curve with equation
- $$y = (2 + x)^3 - (2 - x)^3$$
- has no stationary points. (3)
- (Total 10 marks)**

Q9.

The curve with equation $y = x^3 - 5x^2 + 7x - 3$ is sketched below.



The curve touches the x -axis at the point $A (1, 0)$ and cuts the x -axis at the point B .

- (a) (i) Use the factor theorem to show that $x - 3$ is a factor of

$$p(x) = x^3 - 5x^2 + 7x - 3$$

(2)

- (ii) Hence find the coordinates of B .

(1)

- (b) The point M , shown on the diagram, is a minimum point of the curve with equation $y = x^3 - 5x^2 + 7x - 3$.

- (i) Find $\frac{dy}{dx}$.

(2)

- (ii) Hence determine the x -coordinate of M .

(3)

- (c) Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

(2)

- (d) (i) Find $\int (x^3 - 5x^2 + 7x - 3) dx$.

(4)

- (ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)

(Total 18 marks)

Q10.

The cubic polynomial $p(x)$ is given by $p(x) = (x - 2)(x^2 + x + 3)$.

- (a) Show that $p(x)$ can be written in the form $x^3 + ax^2 + bx - 6$, where a and b are constants whose values are to be found.

(2)

- (b) Prove that the equation $(x - 2)(x^2 + x + 3) = 0$ has only one real root and state its value.

(3)

(Total 5 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.